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High Speed SAR-ADC

ECEN 607 course project

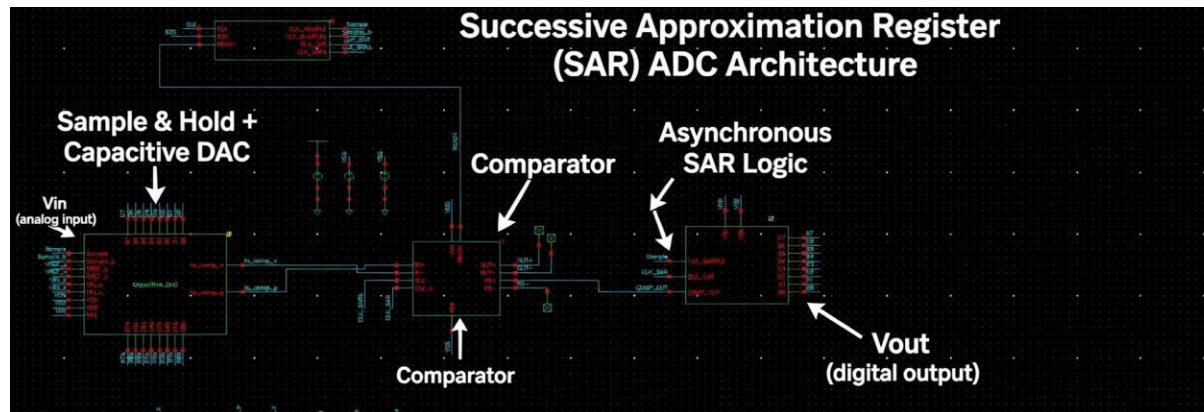
Sai Tharun Sajjanapu

Sri Harsha Darbhasayanam

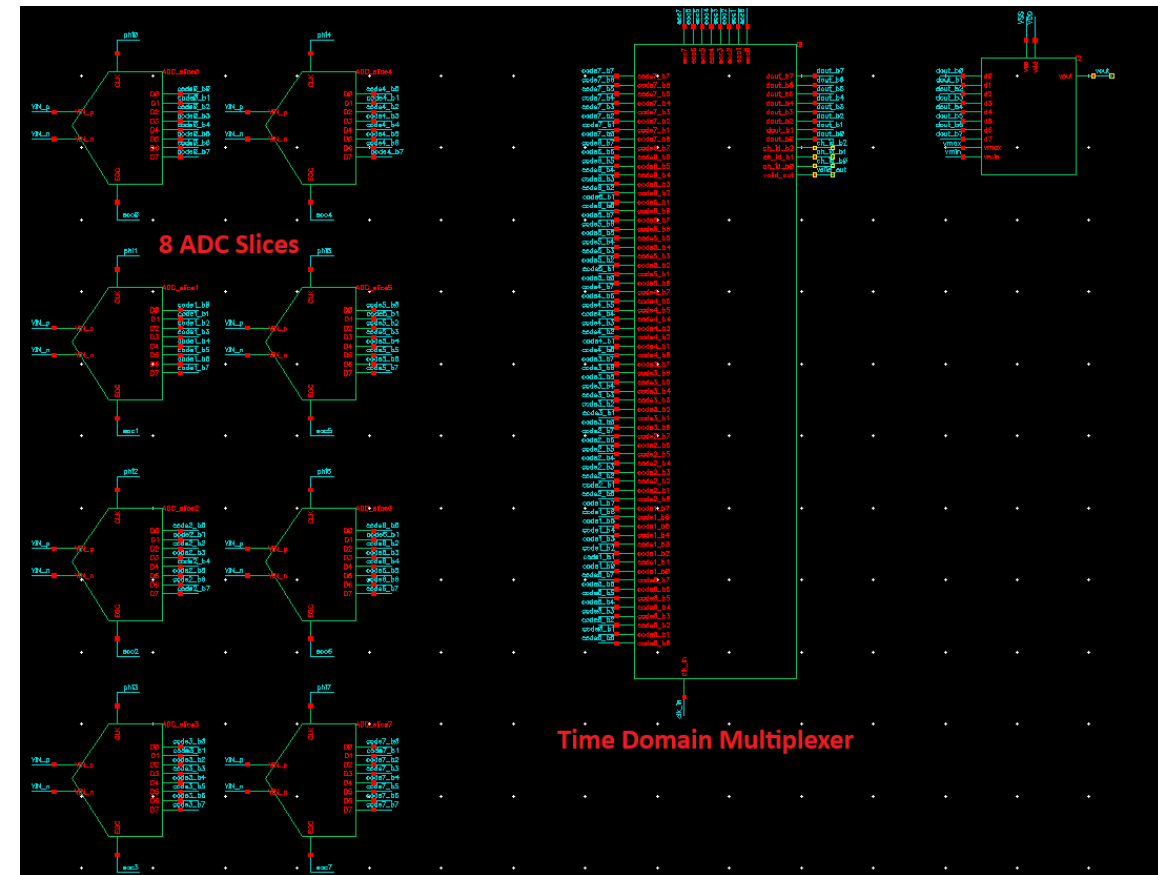
Sheena Selvakumar

Architecture

- Time Interleaving Architecture with 8 slices of unit ADC
- Each slice is an Asynchronous SAR-ADC
- Sampling Rate: 500 MS/s (62.5 MHz per slice)
- VFS = 1.2V, 1LSB = 4.68 mV
- Transistor level: S/H, DAC, switches, comparator
- Verilog-A: TDM, SAR logic



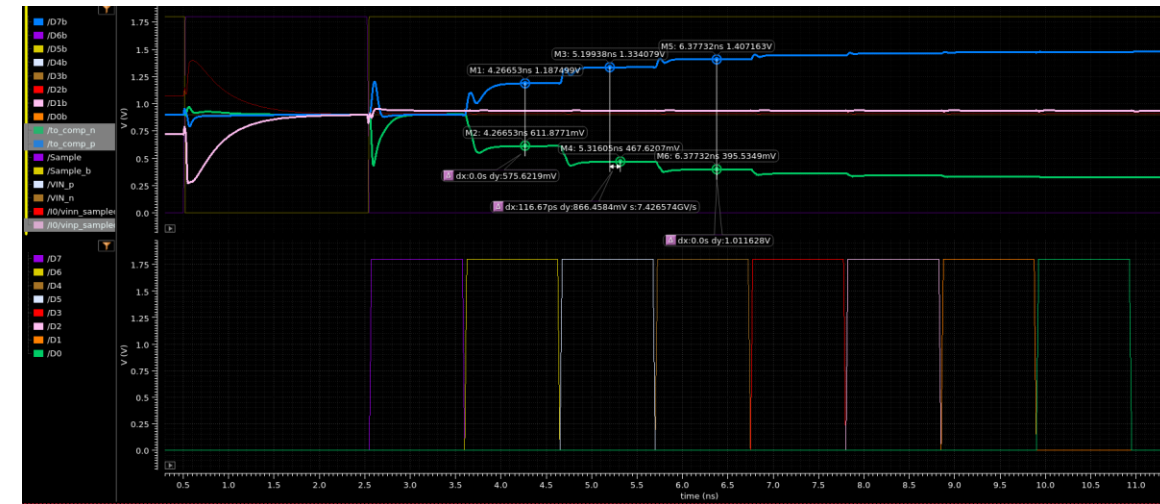
Asynchronous SAR-ADC (unit ADC per slice)



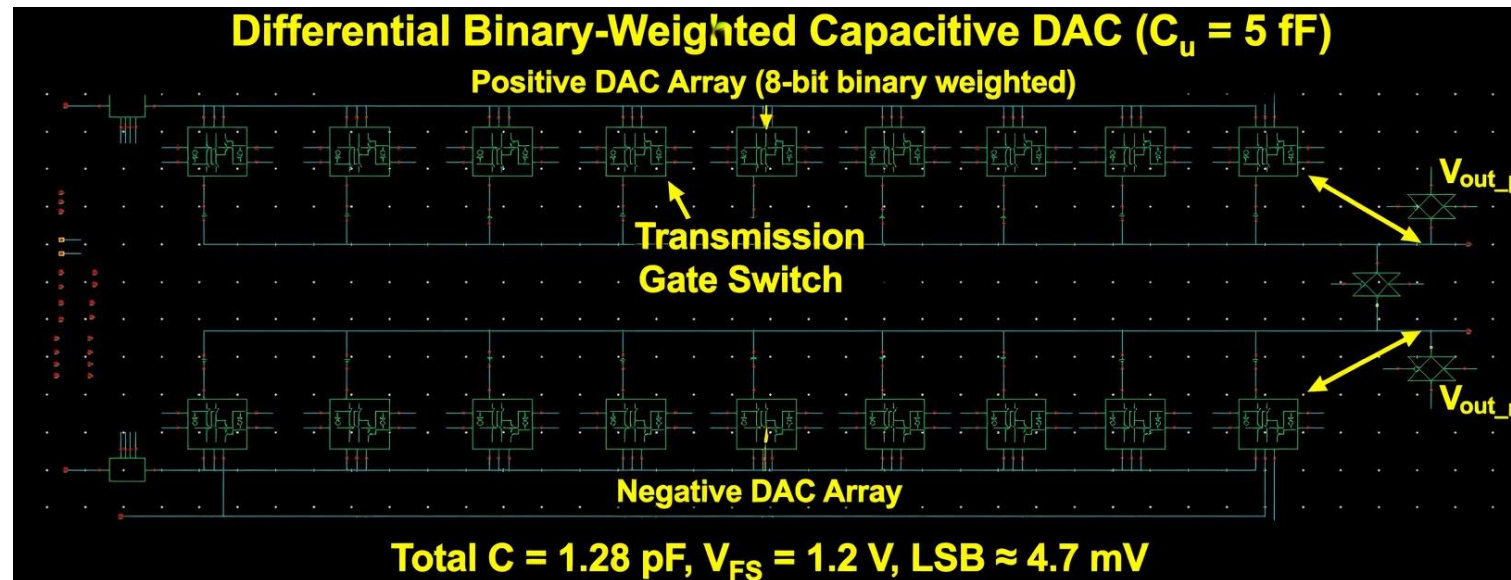
Time Interleaving SAR-ADC

Capacitive DAC

- Fully differential architecture to reject common-mode noise
- Transmission-gate switches for high linearity and reduced charge injection
- Full-scale range $V_{FS} = 1.2\text{ V} \rightarrow \text{LSB} \approx 4.7\text{ mV}$
- Settling performance integrated with S/H:** All bit transitions settle in $< 450\text{ ps}$, meeting the 600 ps SAR comparison budget



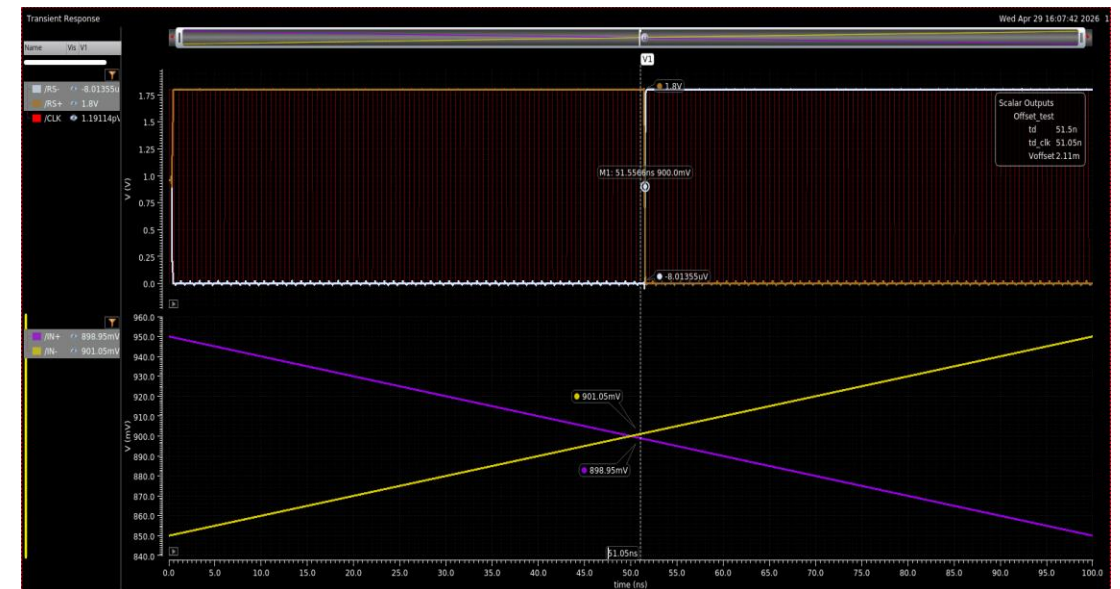
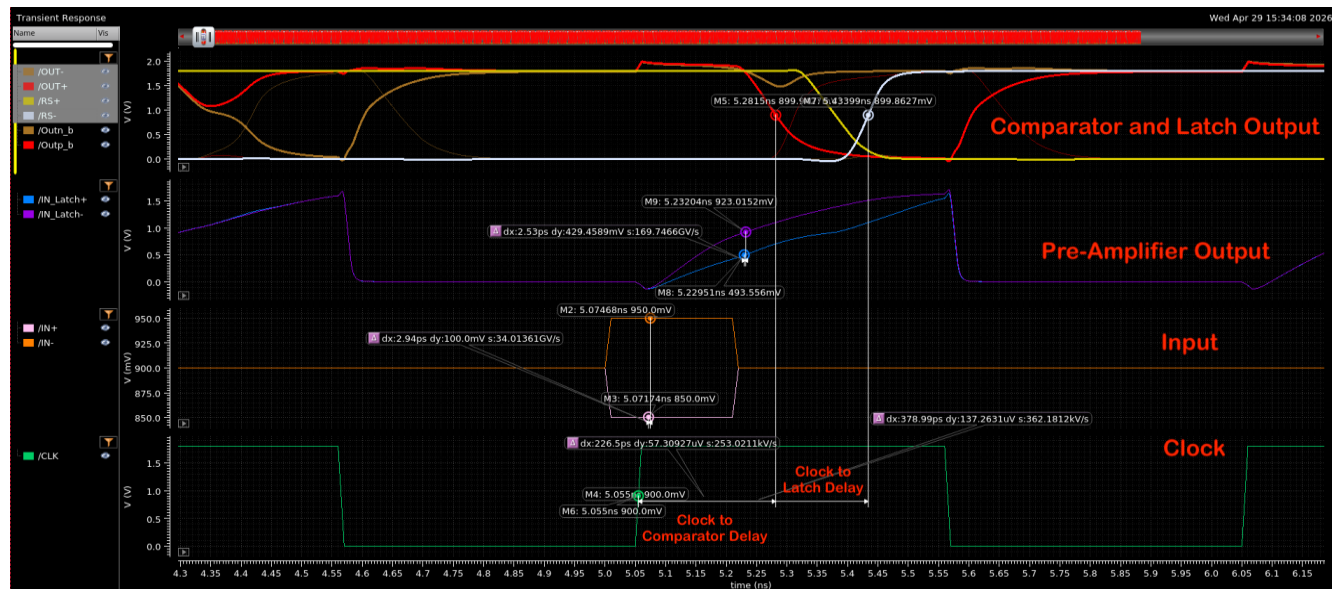
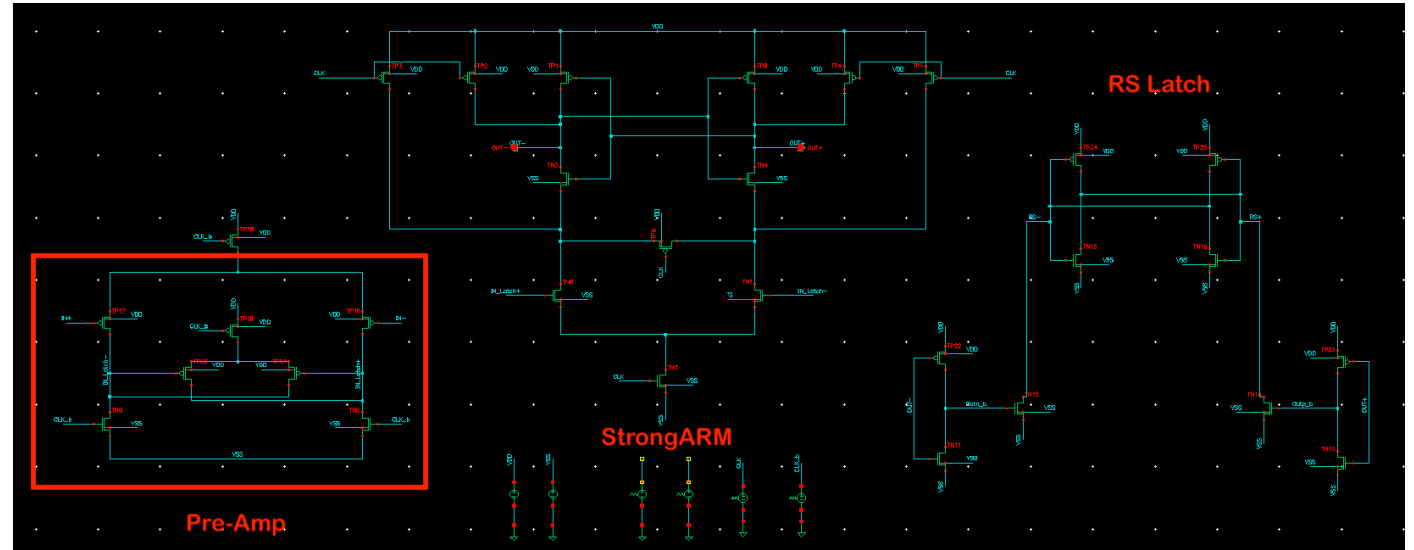
Cap DAC settling during SAR logic



Differential Capacitive DAC

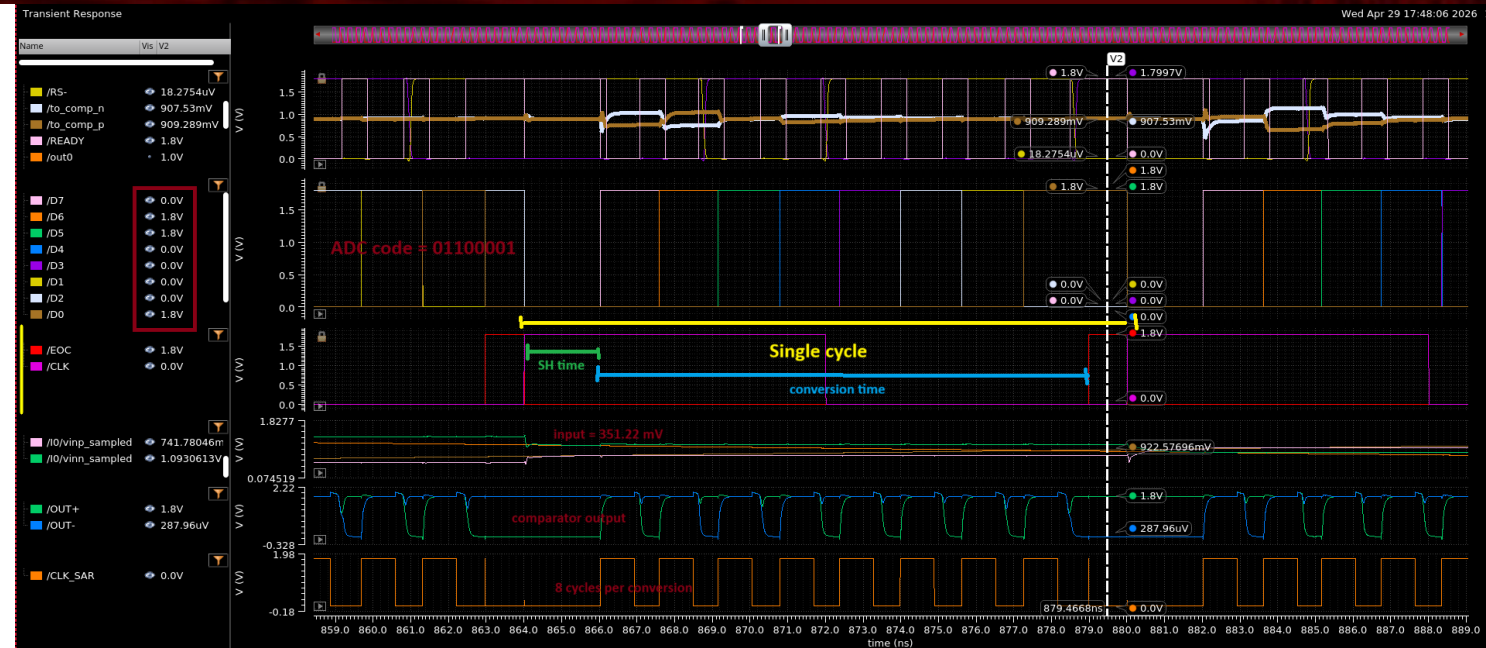
Comparator

- StrongARM Comparator + $\times 4$ Pre-Amplifier
- Raises 100 mV differential input to ≈ 450 mV $\rightarrow$ reduces kickback & metastability
- Clock-to-comparator delay: 230 ps (tested at 1 GHz)
- **Systematic offset:** 2.11 mV (< 0.5 LSB)
- **Monte-Carlo (100 runs):** $\sigma = 3.28$ mV, $\mu = 2.28$ mV

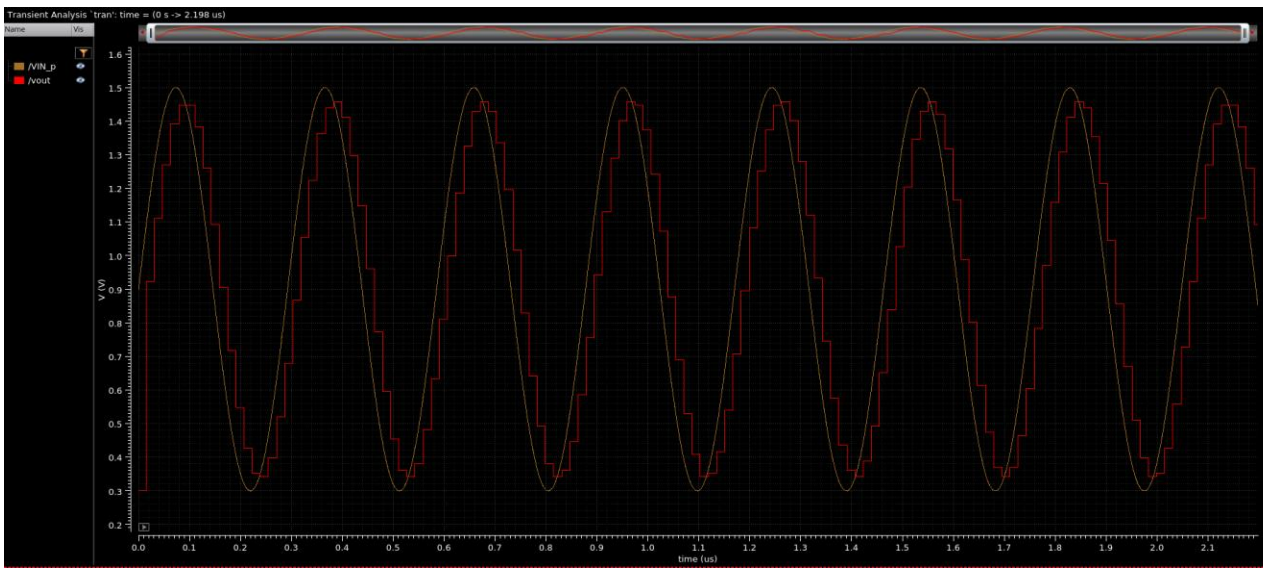


Single Slice ADC

- Runs at 62.5 MS/s (fs/8) with full conversion in 16 ns window
- Integrates bootstrapped S/H + differential cap-DAC + StrongARM comparator + Verilog-A SAR logic
- Power consumption: ≈ 1 mW per slice ($6.047 \text{ mA} \times 1.8 \text{ V}$)

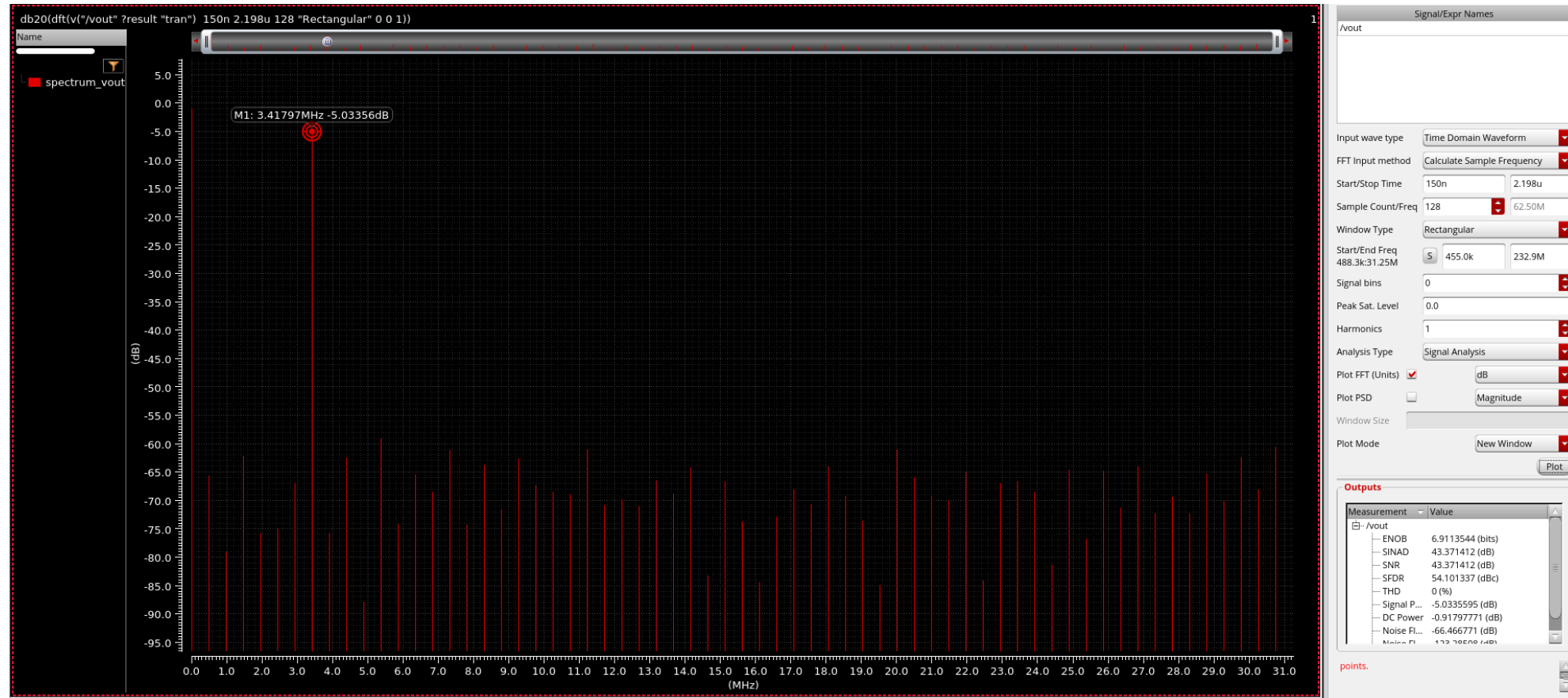


Single conversion cycle of a unit ADC



→ Final ADC output tracking the input sinusoid

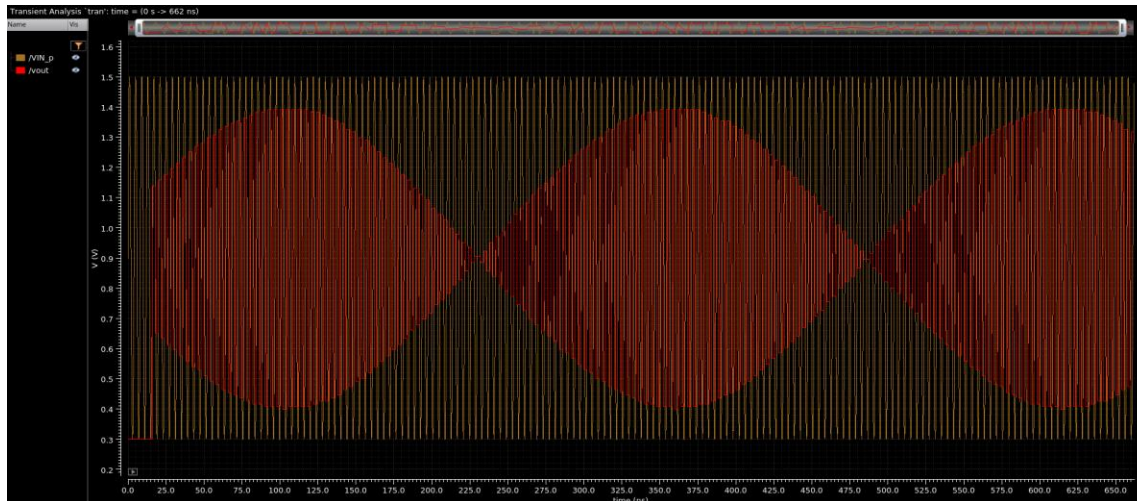
Single Slice ADC FFT



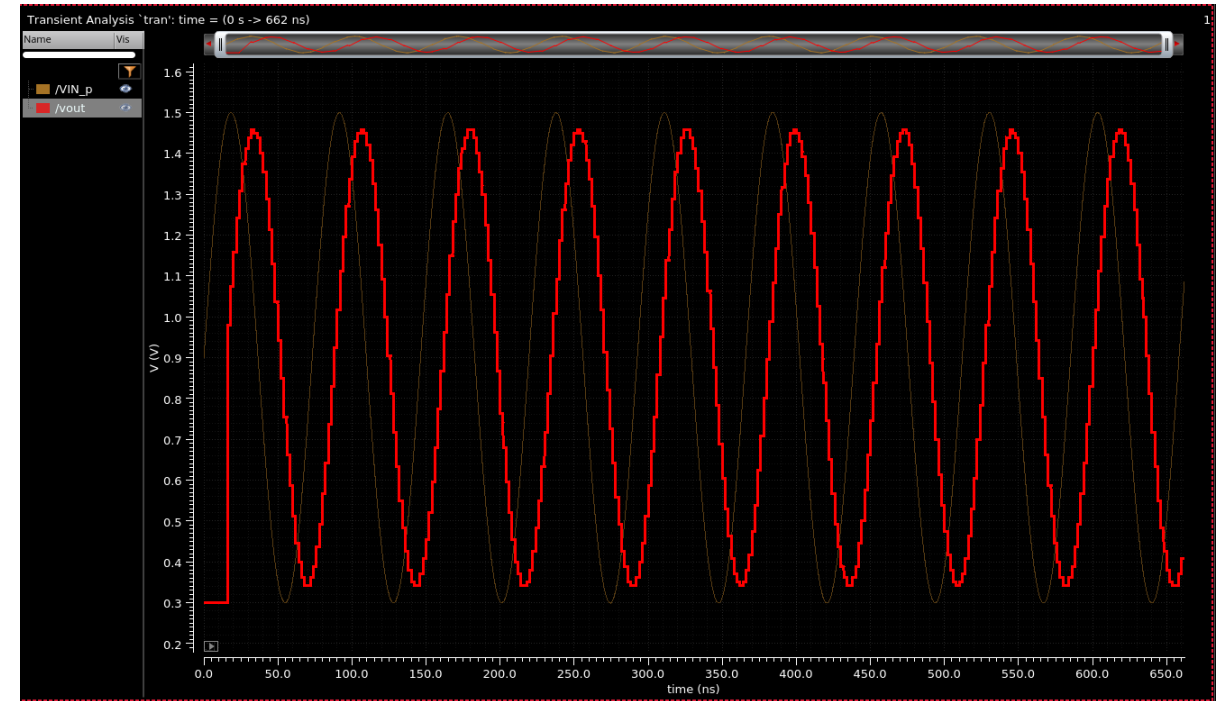
128-point FFT ($f_{\text{in}}(7/128) = 3.42 \text{ MHz}$): ENOB = 6.91 bits, SNR = 43.37 dB

Integrated ADC

- 8-way Time-Interleaved Asynchronous SAR ADC Full system (8 slices + Verilog-A TDM) at fclk = 500 MHz
→ 500 MS/s sampling rate, 250 MHz signal bandwidth
- Schreier FoM = 148.49 dB at ≈ 8 mW total power

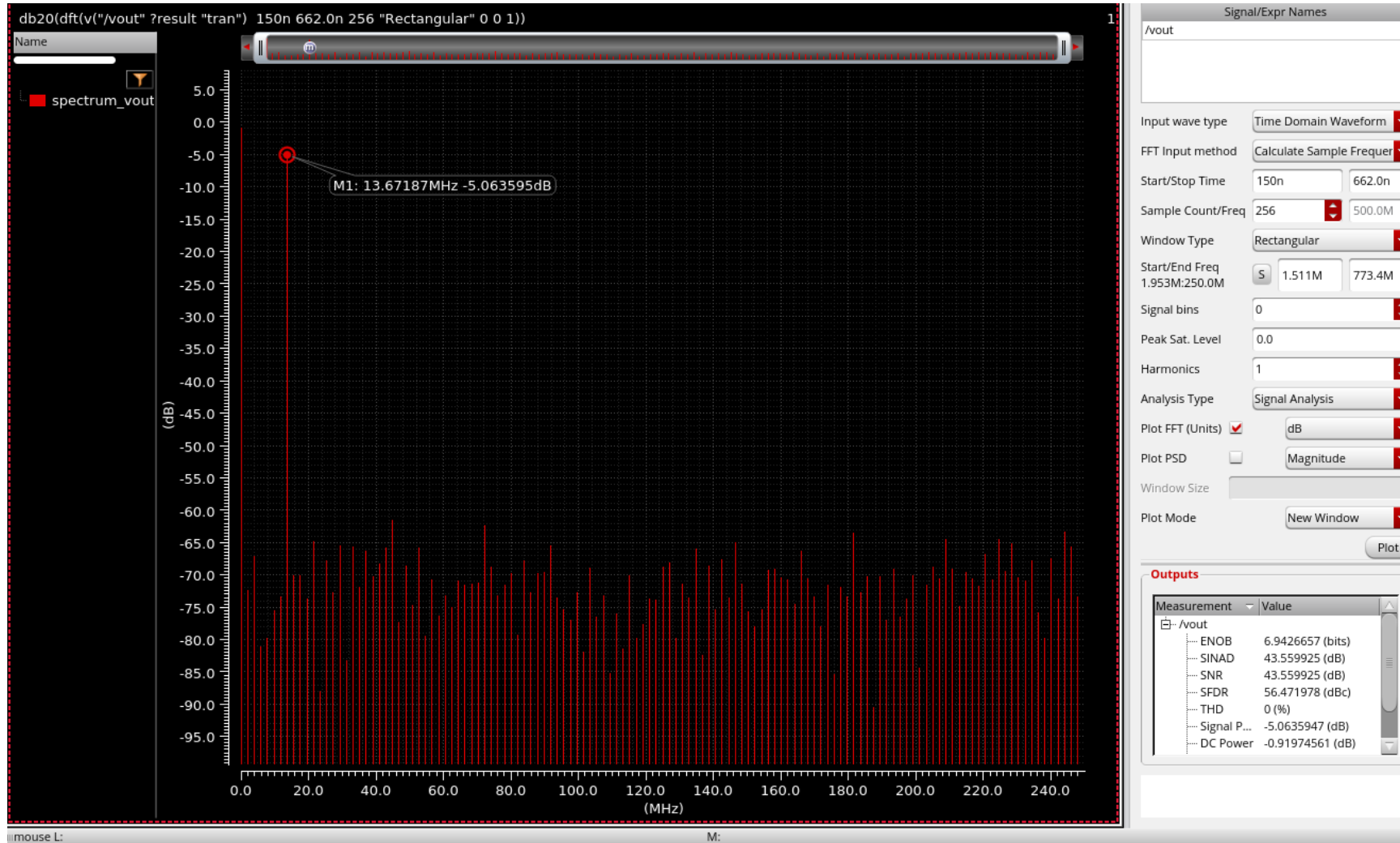


ADC output ($f_{in} = 250\text{MHz}$)



ADC output ($f_{in} = 13.6\text{ MHz}$)

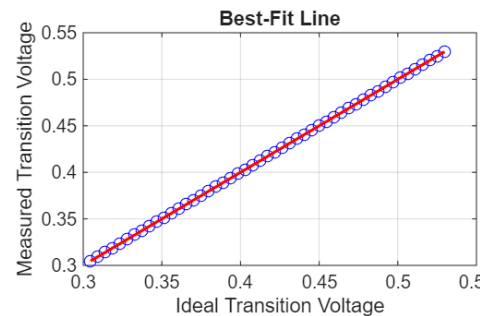
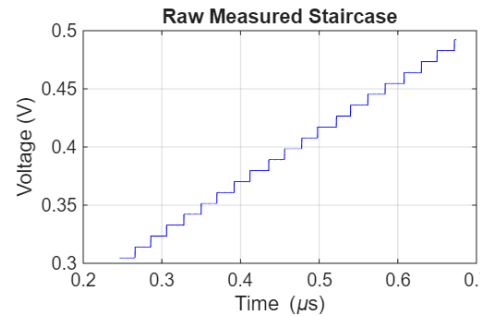
Integrated ADC FFT



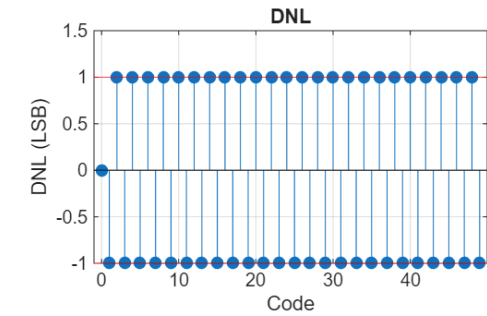
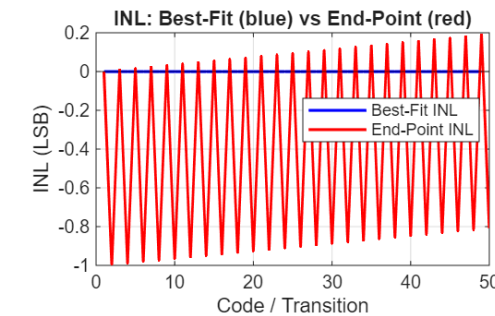
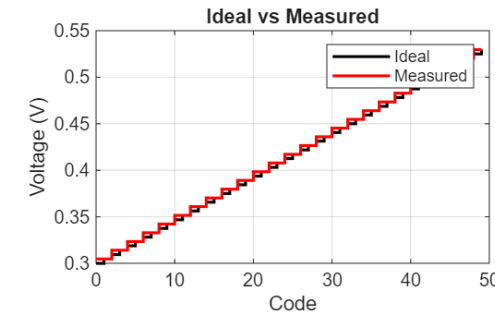
256-point FFT ($f_{in} = 13.6$ MHz): ENOB = 6.94 bits, SNR = 43.55 dB, SFDR = 56.47 dBc

Integrated ADC characteristics

- Input was ramped up and results were used for post processing in matlab
- Observed Behavior: The 8-bit ADC shows perfect Best-Fit linearity (INL ≈ 0 LSB, offset/gain error = 0) but significant endpoint errors:
- DNL alternates between +1 LSB and -1 LSB (peak DNL = 1.000 LSB)
- Endpoint INL displays a classic sawtooth waveform with peak ≈ 0.996 LSB
- **Root Cause:** LSB (bit 0) decision is stuck because the combined errors exceed 1 LSB (≈ 4.688 mV):
 - Sum of noise > 1 LSB
 - kT/C noise on 5 fF unit capacitor
 - Systematic comparator offset (2.11 mV)
 - CDAC settling error
- **Consequence:** Only odd codes (1, 3, 5, ...) are produced $\rightarrow$ missing even codes $\rightarrow$ Effective resolution drops to ~ 7 bits (ENOB degraded) despite the 8-bit architecture



INL/DNL Analysis with End-Point INL



===== SUMMARY =====
 Resolution = 8 bits
 LSB = 4.688 mV
 --- Static Errors (Best-Fit) ---
 Offset Error = -0.00000 V
 Gain Error = 0.0000 % (0 ppm)
 --- Linearity ---
 Peak DNL = 1.0000 LSB
 Peak INL (Best-Fit) = 0.0000 LSB
 Peak INL (End-Point) = 0.9961 LSB

Performance Summary

8-bit SAR ADC | 180 nm CMOS Technology

Parameter	Target	Achieved
Signal BW	250 MHz	250 MHz
Sampling Rate	500 MS/s	500 MS/s
Resolution	8 bit	8 bit
ENOB	8 bits	6.94 bits
SNR	—	43.55 dB
SFDR	—	56.47 dBc
Power	—	≈8 mW
FoM	> 170 dB	148.49 dB



THANK YOU

Any Questions?



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Back up

FSM for Asynchronous clock block

